Zhongsheng Wang

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Summary/Objective

I am currently a second-year PhD student in Computer Science at the University of Auckland, with a research focus on constructing controllable Multimodal Large Language Models (MLLMs). My current research focuses on self-evolving agents and the development of trustworthy multi-agent systems, with a commitment to making practical and impactful contributions to the field of AI.

Education

The University of Auckland — **Auckland, New Zealand**
Doctor of Philosophy in Computer Science (Supervisors: Prof. Jiamou Liu and Dr. Qian Liu) — *2025.04 - 2026.03*

The University of Auckland — **Auckland, New Zealand**
Master of Data Science (Supervisor: Prof. Jiamou Liu) — *2022.07 - 2024.09*

Southwest University — **Chongqing, China**
Bachelor of Engineering in Computer Science and Technologies (Advisor: Prof. Li Li) — *2019.09 - 2023.06*

Research Interests

- Multimodal Large Language Models (MLLMs)
- Self-evolving AI agents
- Multi-agent systems

Project Experience

BedaiaTi.ai: Agentic AI System for Culturally Grounded Storybook Generation — 2025.04 – 2025.07

- Led the core development of a multi-agent AIGC system for Arabic-context children's storybook generation, coordinating specialized agents for story generation, page-level storyboard planning, character-consistent illustration generation, and language localization.
- Designed an asynchronous task orchestration workflow based on message queues and developed a reusable model execution control layer, or *model harness*, to support task scheduling, output validation, logging, and real-time progress feedback.
- Delivered and deployed the system at bedaiati.ai, enabling an end-to-end user workflow from theme selection and story input to complete storybook generation for the Arabic-speaking market.

Industry Experience

AI Engineer — 2024.05 – 2024.08

Atom Intelligence — *Remote*

- Built a Chinese-English CRM speech recognition system based on Whisper and LoRA fine-tuning, covering domain adaptation, model deployment, and business system integration; reduced key error-rate metrics by approximately 30% relative to the previous method.
- Designed an automatic annotation and cleaning workflow for industrial speech data, building a reusable ASR data processing pipeline for training data construction, quality filtering, and iterative model improvement.
- Participated in experimental validation and deployment optimization of the weakly supervised ASR system, connecting data construction, model fine-tuning, and deployment feedback into a closed-loop workflow for real CRM scenarios.

Data/AI Scientist, Summer Intern — 2023.11 – 2024.01

HouGarden Co, Ltd. — *New Zealand*

- Built a property-level issue management database for a real estate chatbot by organizing property information, user questions, and question-answering data, providing a structured knowledge base for industry-specific QA systems.
- Constructed Chinese-English bilingual corpora and a domain terminology database for real estate, and fine-tuned, adapted, and deployed a large language model for official website property information translation, improving stylistic consistency in domain-specific text generation.

Teaching & Academic Talks

Graduate Teaching Assistant **2023.07 – Present**
The University of Auckland *Auckland, New Zealand*

- Supported undergraduate and postgraduate computer science courses by answering student questions, marking assignments and tests, and providing feedback on programming, artificial intelligence, and mathematical reasoning tasks.

- Assisted with courses across artificial intelligence, intelligent agents, software engineering theory, and mathematics for computer science.
- **Selected courses:** COMPSCI 767 Intelligent Software Agents (2026 S1); COMPSCI 713 AI Fundamentals (2026 S1); COMPSCI 367 Artificial Intelligence (2023 S2, 2025 S2); COMPSCI 761 Advanced Topics in Artificial Intelligence (2024 S2); SOFTENG 282 Software Engineering Theory (2025 S1); COMPSCI 120 Mathematics for Computer Science (2025 S1).

Guest Lectures and Invited Talk **2024 – 2026**

- **Text-to-Image Diffusion Models: Limits and Control**, guest lecture for COMPSCI 714 AI Architecture and Design, University of Auckland, May 2026.
- **Reinforcement Learning for LLMs: From Post-Training to Skills and Agentic Systems**, guest lecture for COMPSCI 713 AI Fundamentals, University of Auckland, May 2026.
- **Creative Intelligence: Applications of Large Language Models in Data Generation and Reasoning**, invited talk, University of Electronic Science and Technology of China, Chengdu, China, December 2024.

Academic Services

Conference Reviewer: ACM MM 2026; ICONIP 2026; ACL Rolling Review 2026 January; AAAI 2025; ACM MM 2025; COLM 2025; IJCNN 2025; ICONIP 2025; ACL Rolling Review 2025 May, July, October; IJCNN 2024; ICONIP 2024.

Journal Reviewer: IEEE Transactions on Fuzzy Systems, 2025.

Workshop Reviewer: Reasoning and Planning for Large Language Models @ ICLR 2025.

Awards & Honors

Best Paper Award, AAMAS, 2026.

Best Multimedia Award, ACM Multimedia Asia, 2025.

Master's Dissertation in First-Class, University of Auckland, 2024.

University-level Scholarship, Southwest University, 2020, 2021, 2022.

Publications

Conference & Journal Papers

*** means equal contribution**

Lin Z, **Wang Z**, Liu Q, Zhang X, and Liu J. Narratology Meets Text-to-Image: A Survey of Consistency in AI Generated Storybook Illustrations. *Artificial Intelligence Review*, 2026.

Li M*, **Wang Z***, Li H, and Liu J. R-Debater: Retrieval-Augmented Debate Generation through Argumentative Memory. Accepted by *The 25th International Conference on Autonomous Agents and Multiagent Systems*, AAMAS 2026. **Best Paper Award**

Deng S, **Wang Z**, Mao R, Ciprian D G, and Liu J. CLER: Improving Multimodal Financial Reasoning by Cross-MLLM Error Reflection. Accepted by *The 40th AAAI Conference on Artificial Intelligence*, AAAI 2025.

Li X, Ni L, Wang X, Tang Y, Li R, Liu J, and **Wang Z**. LLM-based Business Process Models Generation from Textual Descriptions. Accepted by *The 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, IJCNLP-AACL 2025.

Wang Z, Lin M, Lin Z, Shakib Y, Liu Q, and Liu J. CharCom: Composable Identity Control for Multi-Character Story Illustration. Accepted by *The 7th ACM International Conference on Multimedia in Asia*, ACM MM Asia 2025. **Best Multimedia Award**

Wang Z, Wang S, Wang J, Liang Y, Zhang Y, and Liu J. Weak Supervision Techniques towards Enhanced ASR Models in Industry-level CRM Systems. *International Conference on Neural Information Processing*, ICONIP 2024.

Wang Z, Liu J, Bao Q, Rong H, and Zhang J. ChatLogic: Integrating Logic Programming with Large Language Models for Multi-Step Reasoning. *International Joint Conference on Neural Networks*, IJCNN 2024.

Qi Q, Ni L, **Wang Z**, Zhang L, Liu J, and Witbrock M. Epic-Level Text Generation with LLM through Auto-Prompted Reinforcement Learning. *International Joint Conference on Neural Networks*, IJCNN 2024.

Preprint & Workshop Papers

Xiao X, Shen S, Bao Q, Rong H, Liu K, **Wang Z**, and Liu J. CoRA: Optimizing Low-Rank Adaptation with Common Subspace of Large Language Models. Arxiv pre-print